

Technology Stack

Date	08 October 2026
Team ID	UNM1611BHAVANA
Project Name	AI FitTrack API
Maximum Marks	2 Marks

Define Your Technology Stack:

The technologies below were chosen so that the AI FitTrack API is scalable, maintainable and aligned with the project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	None (API-only backend); tested with Postman	The project delivers a REST API. Any web or mobile client can consume it, so no UI layer is built.
2	Backend / Server-Side	Node.js with Express.js	Fast, non-blocking I/O suited to API workloads; one language (JavaScript); large npm ecosystem; team familiarity. JWT, bcrypt, dotenv and a validation library are used for security and configuration.
3	Database / Data Storage	MongoDB with Mongoose	Flexible document model fits varied workout and meal logs; Mongoose adds schemas, validation and easy queries.
4	Cloud / Hosting / Deployment	Local Node.js environment (cloud hosting such as Render or AWS can be added later)	Runs locally for evaluation at no cost; the stateless design allows easy deployment to a cloud platform.
5	Version Control & CI/CD	Git and GitHub	Shared repository for team collaboration, history and code review; automated test runs can be added with GitHub Actions.
6	Third-Party APIs / Other Tools	AI / LLM recommendation service (REST), Postman, Jest and Supertest	The AI service generates personalised workout and diet plans; Postman is used for manual API checks and Jest / Supertest for automated tests.